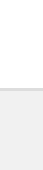


- Airlock rotary valve
- Bag tipping
- Big Bag tipping
- Big Bag filling
- Bin blenders
- Bucket elevators
- Roots blowers
- Butterfly valves
- Bulk mechanical conveying
- Conveying pipes
- Conveying pipe bends
- Cyclones
- Design of a cyclone : Leith and
Licht model
- Dust collection systems
- Filters
- IBC/FIBC
- Load cells
- Magnets (foreign body
prevention)
- Dry mixers
- Paddle mixers
- Pipe diverters
- Ribbon blenders
- Rotary valves
- Screw conveyors
- Screw conveyors design
calculation (capacity)
- Flexible Screw Conveyors
- Shear mixers
- Solids / Liquid Mixers
(suspension)
- V blenders
- Vibrating sieves
- Vibrating tubes

Cyclone design

Step by step design guide and cyclone shortcut calculation tool

Follow us on [Twitter](#) 

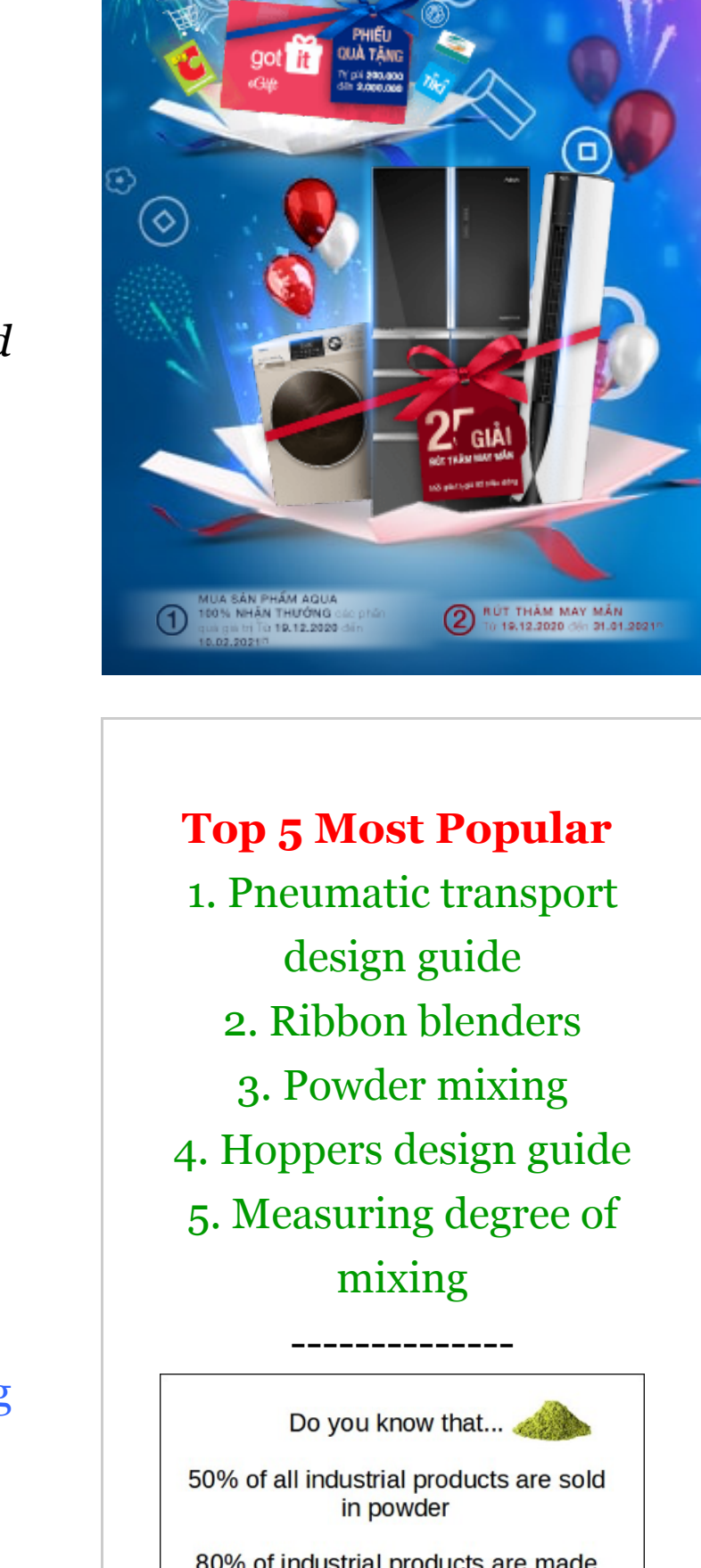
Question, remark ? Contact us at admin@powderprocess.net

Section summary
1. Introduction
2. Applications of cyclones
3. Cyclone Standard Geometry
4. Cyclone Step by Step design guide
5. Cyclone design Excel calculation tool

Fitlaira Products

Design and supply of dust collection and pollution control equipment and spare parts.

fitlaira.com.au



1. Introduction

Cyclones and their design

There are different processes for collecting the dust in a gas stream (see [global overview here](#)). among them, cyclones are probably one of the most widespread solution, in any industry. Cyclone dust collectors are fairly simple from a mechanical point of view and therefore generally provide a cost effective solution. However, assessing the performance of a cyclone and designing a new equipment for a particular application is not always well understood and only partial literature is often found. The objective of this page is to provide a step by step approach to cyclone dust collector design. This can be sufficient to check quickly the performance of an existing cyclone or during pre-design, *one should however reckon that the methodology below is not suited for detailed design which should be carried out with a reputable supplier which will likely have refined the original calculation codes provided in literature and made them more precise*. One should also remark that the method given is only one among several published models which may have different accuracy.

Article in development, please stay tuned for updates

2. Applications of cyclones

Where are used cyclones ?

Cyclones dust collectors are particularly used in the following applications :

- Plastics : after transport of pellets, to catch plastic dust
- Wood industry : to collect dust from sawmills
- Chemicals : to collect dust from a process or at the end of a [pneumatic conveying line](#) to control emissions
- Agriculture : to dedust the air used to convey material to a silo

- Top 5 Most Popular
1. Pneumatic transport design guide

2. Ribbon blenders

3. Powder mixing

4. Hoppers design guide

5. Measuring degree of mixing

Top 5 New

1. Continuous Dry Mixing

2. Mixing speed

3. Mixer cycle time optimization

4. Batch / continuous mixing

5. Energy Savings

3. Cyclones standard geometry

What are the standard dimensions of cyclones ?

Cyclones efficiency is directly related to their geometry, which has been the object of various research. From these research papers, a set of STANDARD dimensions have been defined. **Those dimensions, or rather proportions, constitute the basis of most of the design across the industry**. It is recommended to keep those standard configurations, or some adaptation by reputable suppliers, and not modify it. Specific design can still be developed for specific high value applications (FCC for example) but it goes beyond the methodology presented here, requiring modelization, pilot trials,...etc...

The table below is due to Koch and Licht (1977) and is summarizing the work of different authors (Lapple, Stairmand...)

Dimensions	Standard Geometries for cyclones with tangential inlet				
	Standard		High efficiency		
	Lapple	Swift	Peterson Whitby	Stairmand	Swift
H/D _c = K _H	0.5	0.5	0.583	0.5	0.44
B _i /D _c = K _B	0.25	0.25	0.208	0.2	0.21
S _i /D _c = K _S	0.625	0.6	0.583	0.5	0.5
D _i /D _c = K _i	0.5	0.5	0.5	0.5	0.4
L/D _c = K _L	2	1.75	1.333	1.5	1.4
Z _i /D _c = K _Z	2	2	1.84	2.5	2.5
D _e /D _c = K _e	0.25	0.4	0.5	0.375	0.4

Table 1 : Standard cyclone geometries for a tangential inlet

All the dimensions of the cyclones are related to the diameter Dc. A standard geometry is then selected and the diameter Dc is adjusted to get the desired performance.

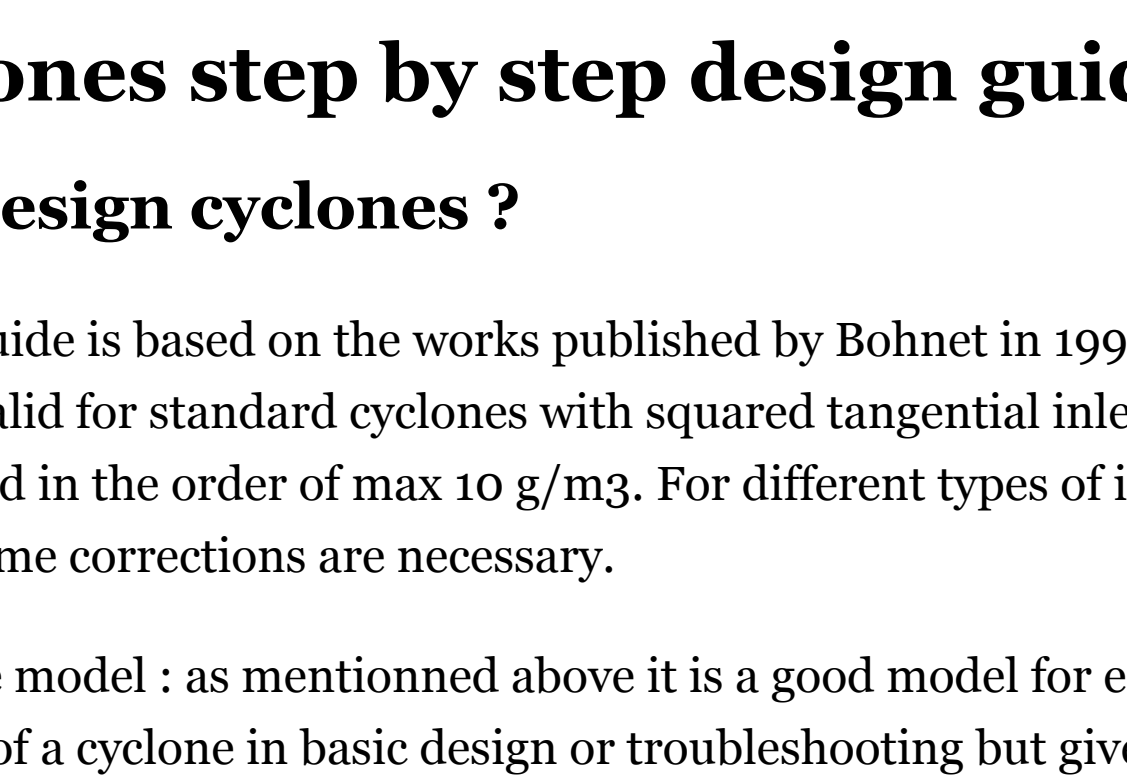


Figure 1 : Cyclone drawing and nomenclature of characteristic geometry

4. Cyclones step by step design guide

How to design cyclones ?

This design guide is based on the works published by Bohnet in 1997. The approach is valid for standard cyclones with squared tangential inlets and with a small dust load in the order of max 10 g/m3. For different types of inlet or higher dust loads, some corrections are necessary.

Validity of the model : as mentioned above it is a good model for estimating the performance of a cyclone in basic design or troubleshooting but gives errors up to 40% vs experiments, depending on the conditions, thus detail calculations should be done with the help of a company specializing in cyclone design and having improved the calculation code.

4.1 Calculate K ratios

If you design a new cyclone, chose one of the standard geometry in table 1 and assume a diameter Dc. If you test an existing cyclone, determine the different ratios for the actual equipment you are evaluating.

K ratios : KH, KB, KS, Ki, KL, KZ, KP from table 1 or actual cyclone dimension

4.2 Calculate the following geometrical dimensions

$$\frac{A_e}{A_i} = \frac{4}{\pi} \frac{K_B K_H}{K_i^2}$$
$$\frac{R_i}{r_e} = \frac{K_i}{1 - K_B}$$
$$\frac{A_e}{A_i} \frac{R_i}{r_e} = \frac{4}{\pi} \frac{K_H}{K_i} \frac{K_B}{1 - K_B}$$

$$\frac{A_f}{A_i} = -1 + 4 \frac{K_S}{K_i} + \frac{1}{K_i} [1 + 4 K_L + 2 K_Z (1 + K_S)]$$

- With :
- A_e = product inlet section area (m2)
 - A_i = gas outlet section area (m2)
 - R_i = radius of gas outlet pipe (m)
 - r_e = average radius of the fluid vein (m)
 - A_f = area of friction of powder on the sides of the cyclone (m2)
 - K_B = B_c/D_c
 - K_H = H_c/D_c
 - K_i = D_i/D_c
 - K_L = L_c/D_c
 - K_Z = Z_c/D_c
 - K_S = S_c/D_c
 - D_c = diameter of the cyclone (m)

4.3 Calculate inlet and outlet velocities

$$u_{ce} = \frac{\dot{V}_c}{K_B K_H D_c^2}$$
$$u_{ci} = \frac{4 \dot{V}_c}{K_i^2 \pi D_c^2}$$

- With :
- V_c = volumetric flow of the continuous phase (gas) (m3/s)
 - u_{ce} =inlet velocity (m/s)
 - u_{ci} =outlet velocity (m/s)
 - K_i = D_i/D_c
 - K_B = B_c/D_c
 - K_H = H_c/D_c
 - D_c = diameter of the cyclone (m)

4.4 Calculate friction coefficients

$$C_e = 1 - \left(0.680 - 0.151 \frac{K_i^2}{K_B K_H} \right) K_B^{1/3}$$
$$\frac{u_{cc}}{u_{ce}} = (0.889 - 0.408 K_B)^{-1}$$
$$Re_c = \frac{\rho_c u_{cc} D_c}{\mu_c}$$
$$C_f = 2.5 \cdot 10^{-3} + \frac{144}{Re_c}$$

- With :
- C_e = coefficient of contraction at inlet
 - K_i = D_i/D_c
 - K_B = B_c/D_c
 - K_H = H_c/D_c
 - u_{cc} =inlet velocity (m/s)
 - u_{cc} =cyclone walls velocity (m/s)
 - Re_c = Reynolds number
 - μ_c = viscosity of the continuous phase (Pa.s)
 - D_c = diameter of the cyclone (m)
 - ρ_c = density of the continuous phase (kg/m3)
 - C_f = friction coefficient
 - A_f = area of friction of powder on the sides of the cyclone (m2)

4.5 Calculate the characteristics velocities

$$u_{ci} = \frac{\dot{V}_c}{\pi D_c^2 K_i (K_L + K_Z - K_S)}$$
$$\frac{u_{c\theta i}}{u_{ci}} = \left[C_e \frac{A_e}{A_i} \frac{R_i}{r_e} + C_f \frac{A_f}{A_i} \sqrt{\frac{R_i}{r_e}} \right]^{-1}$$

- With :
- u_{ci} = gas velocity at the radius Ri (m/s)
 - V_c = volumetric flow of the continuous phase (gas) (m3/s)
 - K_i = D_i/D_c
 - K_L = L_c/D_c
 - K_Z = Z_c/D_c
 - K_S = S_c/D_c
 - D_c = diameter of the cyclone (m)
 - u_{cθi} = (m/s)
 - u_{ci} =outlet velocity (m/s)
 - C_e = coefficient of contraction at inlet
 - A_e = product inlet section area (m2)
 - A_i = gas outlet section area (m2)
 - R_i = radius of gas outlet pipe (m)
 - r_e = average radius of the fluid vein (m)
 - C_f = friction coefficient
 - A_f = area of friction of powder on the sides of the cyclone (m2)

4.6 Calculate the cut off diameter

Particles having a diameter equal to the cut off diameter are captured with an efficiency of 50%. It means that the cyclone will capture 50% of the particles having this diameter in the gas stream and will let through the other 50%.

$$d_c = 2.846 \sqrt{\frac{u_{ci} \mu_c K_i D_c}{\Delta \rho u_{c\theta i}^2}}$$

- With :
- u_{ci} = gas velocity at the radius Ri (m/s)
 - μ_c = viscosity of the continuous phase (Pa.s)
 - K_i = D_i/D_c
 - D_c = diameter of the cyclone (m)
 - Δρ = difference in densities (kg/m3)
 - u_{cθi} = (m/s)

4.7 Calculate the efficiencies

The efficiencies are calculated relatively to the cut off diameter. Bigger particles will lead better efficiencies. Smaller particles to lower efficiencies. A factor Γ is used in the calculation and is usually in the order of 3 (+/- 1).

$$\eta \left(\frac{d_i}{d_c} \right) = \frac{1}{2} + \frac{1}{2} \cos \left(\pi \left[1 - \frac{\ln \left(\Gamma \frac{d_i}{d_c} \right)}{\ln \Gamma^2} \right] \right) \text{ with } \Gamma \text{ usually } = 3$$

- d_i = particle of diameter i for which the efficiency is calculated (m)
- d_c = cut off diameter (m)

4.8 Calculate the pressure drop

$$\frac{A_f}{\pi R_i \sqrt{Re_c} R_i} = \frac{1}{K_i^{3/2}} [1 - K_i^2 + 4 (K_i K_S + K_L) + 2 K_Z (1 + K_S)]$$
$$\Delta P_c = \xi_c \frac{\rho_c}{2} u_{ci}^2$$
$$\xi_c = \xi_{ce} + \xi_{ci}$$
$$\xi_{ce} = \frac{R_i}{R_c} \left[\left(1 - C_f \frac{A_f}{\pi R_i \sqrt{Re_c} R_i} \frac{u_{c\theta i}}{u_{ci}} \right)^{-2} - 1 \right] \left(\frac{u_{c\theta i}}{u_{ci}} \right)^2$$
$$\xi_{ci} = C_{\beta} \left[2 + 3 \left(\frac{u_{c\theta i}}{u_{ci}} \right)^{4/3} + \left(\frac{u_{c\theta i}}{u_{ci}} \right)^2 \right]$$

- With :
- ΔP_c = cyclone pressure drop (Pa)
 - ξ_c = total pressure drop coefficient of the cyclone
 - ξ_{ce} = pressure drop coefficient in the inlet and inside the cyclone
 - ξ_{ci} = pressure drop coefficient in the outlet of the cyclone
 - C_β = 0.70 to 0.75

5. Cyclone design Excel calculation tool

A [simplified version of the calculation tool](#) can be found [here](#) - a more complete tool will be developped soon. Note that this tool **cannot be used for detail design** as stated in the file, always link with a commercial company to confirm the design.

Sources

Bohnet 1997

Advanced coating

Coating solutions for you

kobelco-coating.com

OPEN